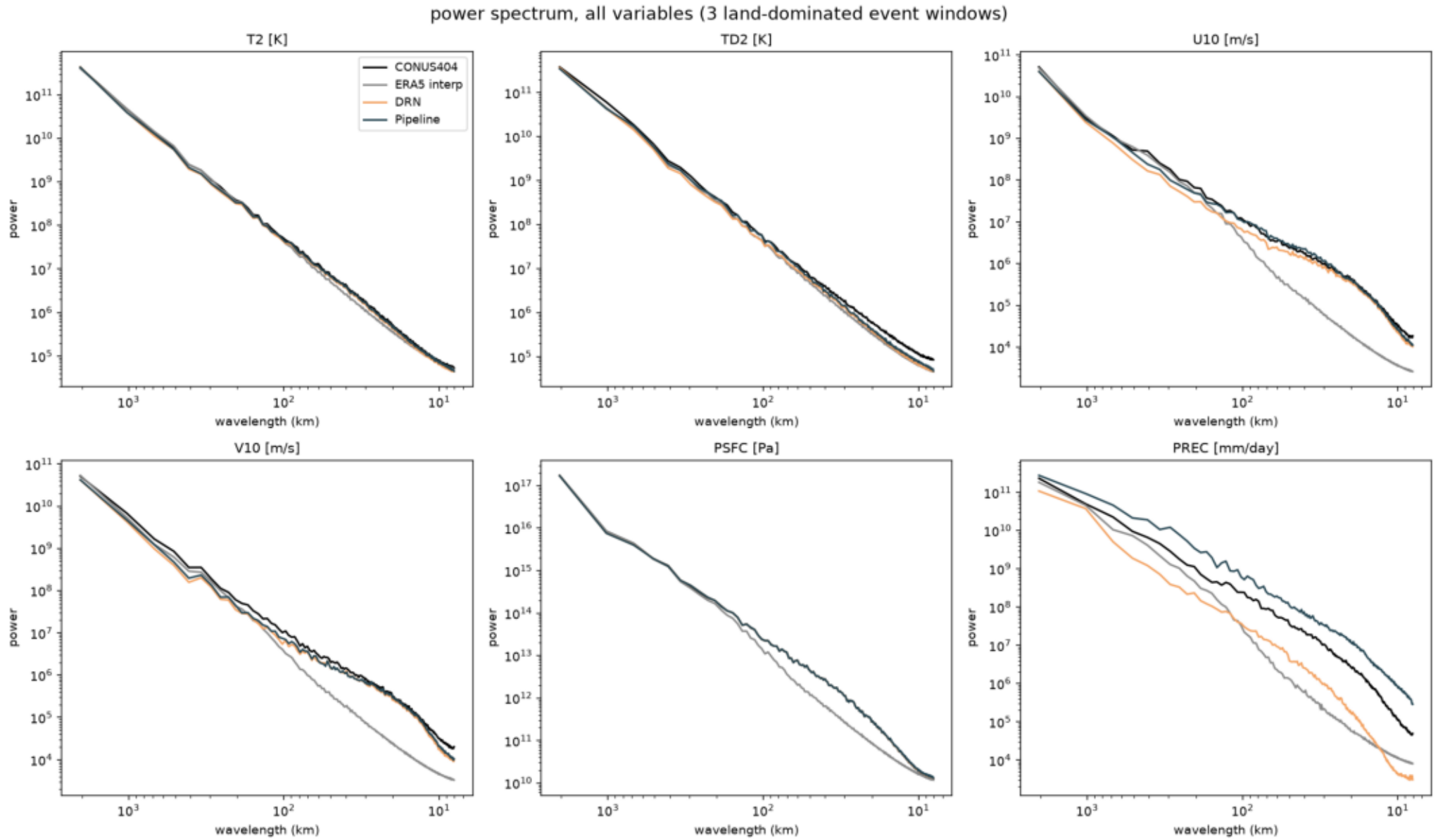
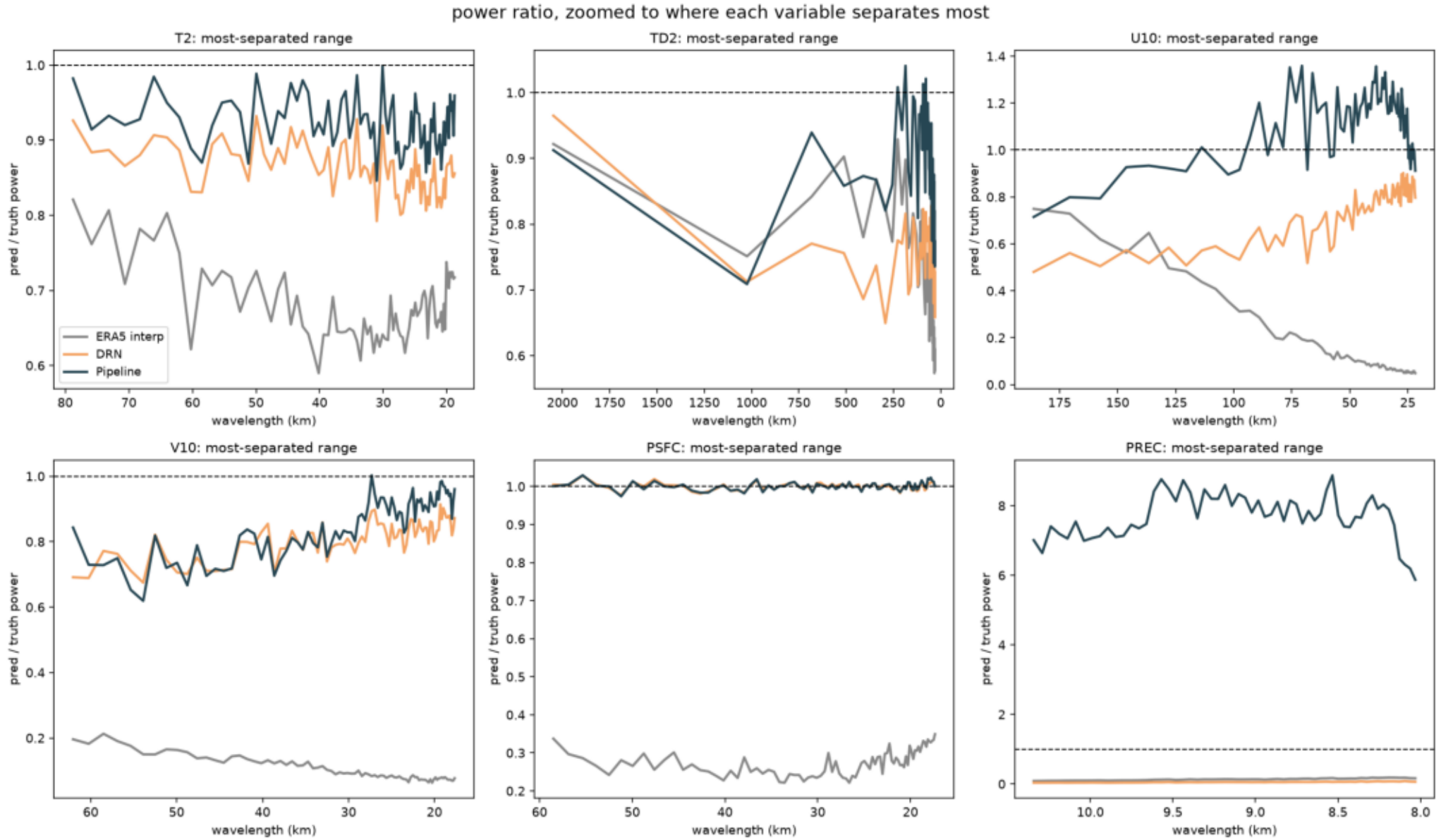
ERA5→CONUS404 Latent CorrDiff

Frequency-domain detail, CorrDiff comparison, Anderson Creek Fire case study

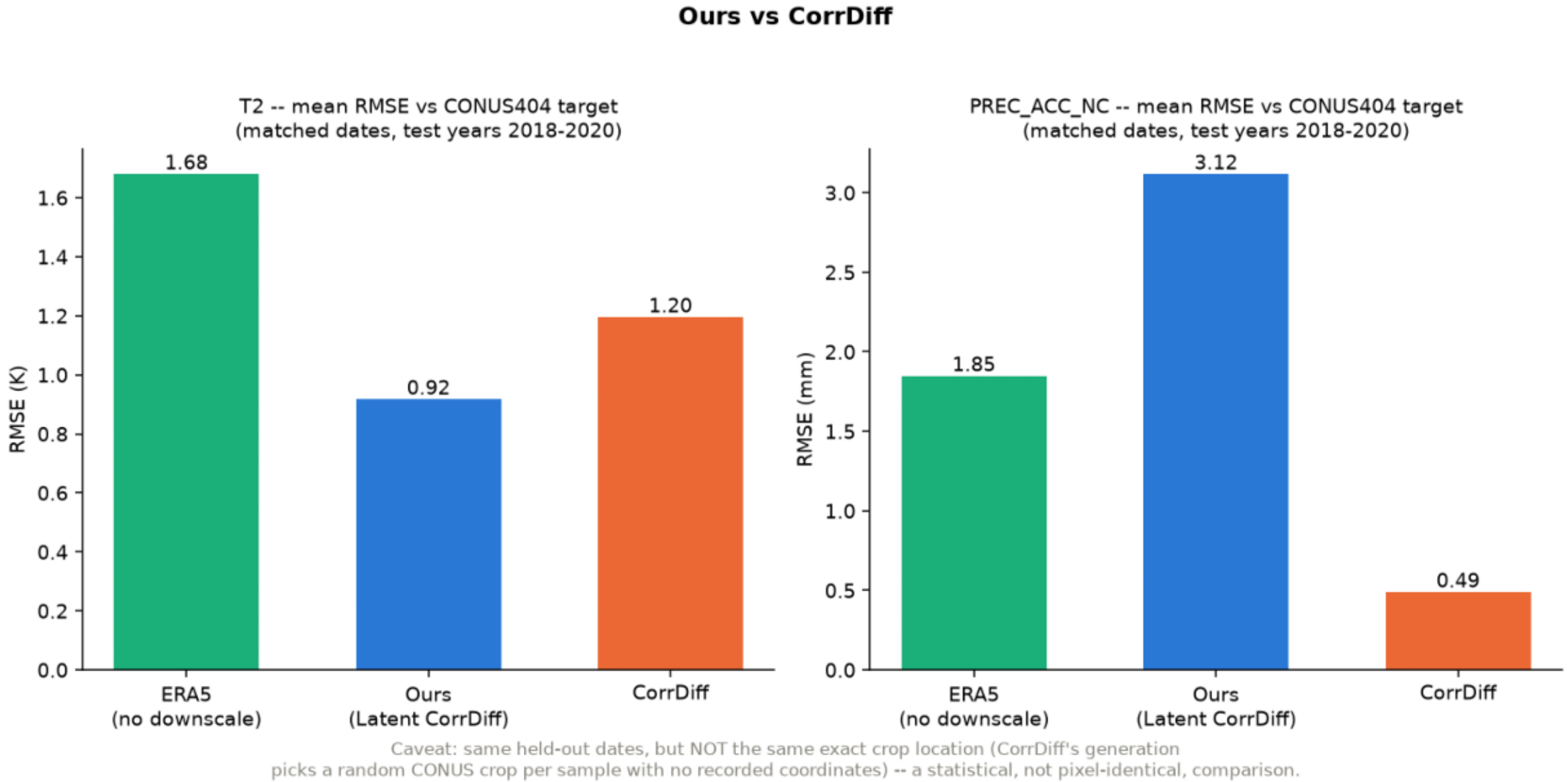
power spectrum, all variables, raw view



power spectrum, all variables, zoomed to where they separate most



ours vs corrdiff, T2 and precip only, mean RMSE



accuracy, all variables, corrdiff vs ours

accuracy, all variables, mean over the 6 matched dates

corrdiff and our crop locations differ for each date, so the two era5 baselines below are not the same patch, do not compare them to each other directly

T2 [K]

corrdiff crop: era5 2.03, pipeline 1.32

our crop: era5 1.56, pipeline 1.30

TD2 [K]

corrdiff crop: era5 2.32, pipeline 1.74

our crop: era5 1.41, pipeline 1.95

U10 [m/s]

corrdiff crop: era5 0.98, pipeline 0.77

our crop: era5 0.96, pipeline 1.01

V10 [m/s]

corrdiff crop: era5 0.98, pipeline 0.76

our crop: era5 0.93, pipeline 1.05

PSFC [Pa]

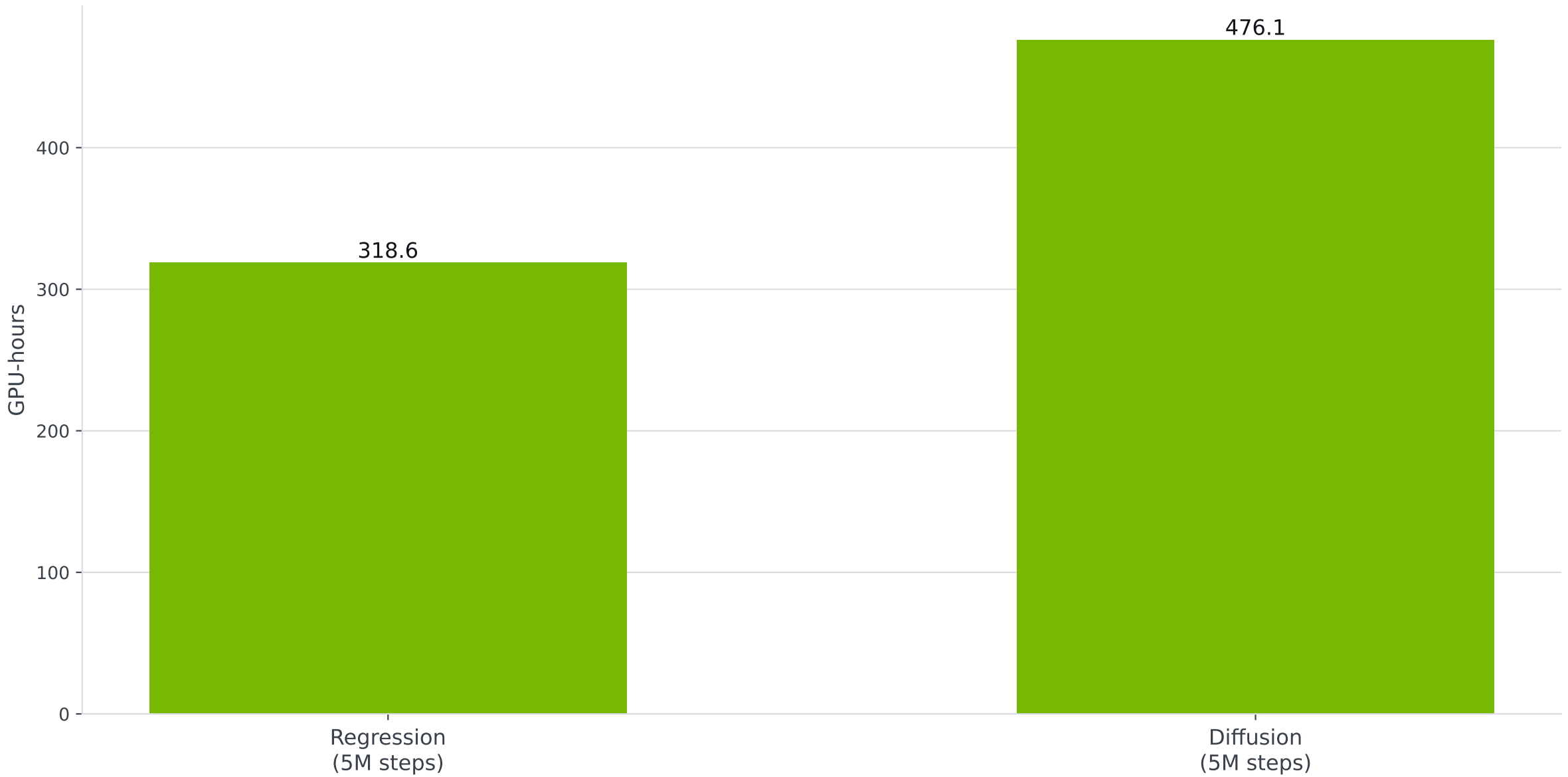
corrdiff crop: era5 991.14, pipeline 121.03

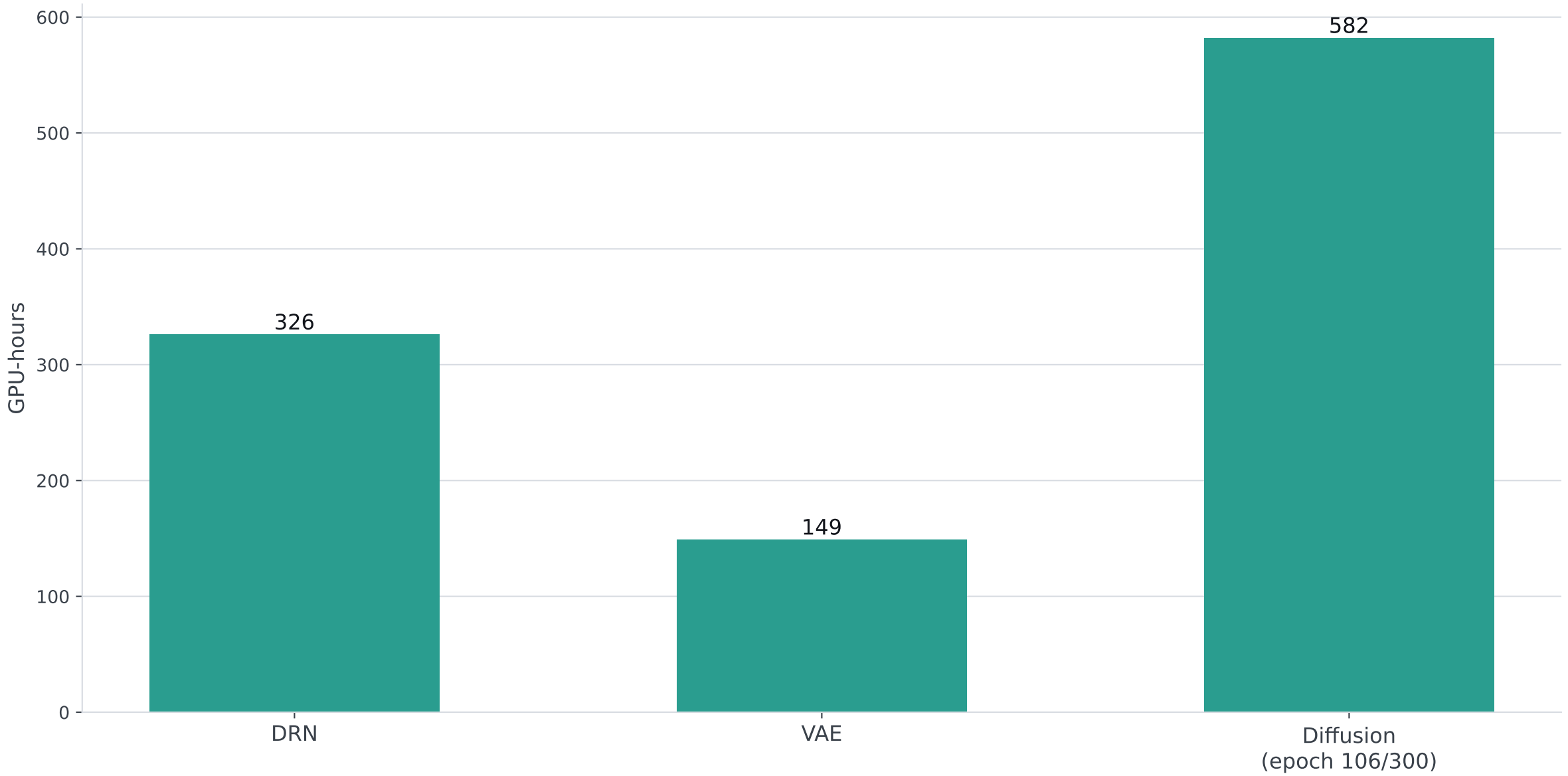
our crop: era5 605.61, pipeline 204.51

PREC_ACC_NC [mm/day]

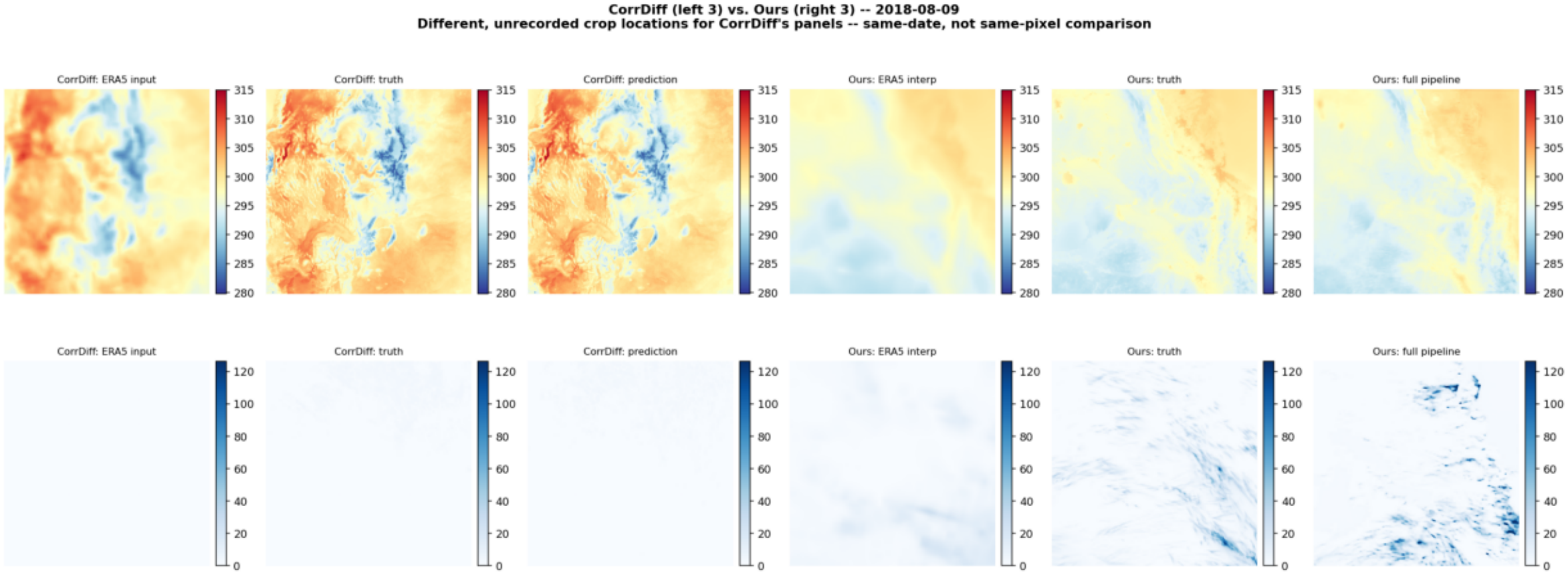
corrdiff crop: era5 0.72, pipeline 0.55

our crop: era5 4.37, pipeline 9.90

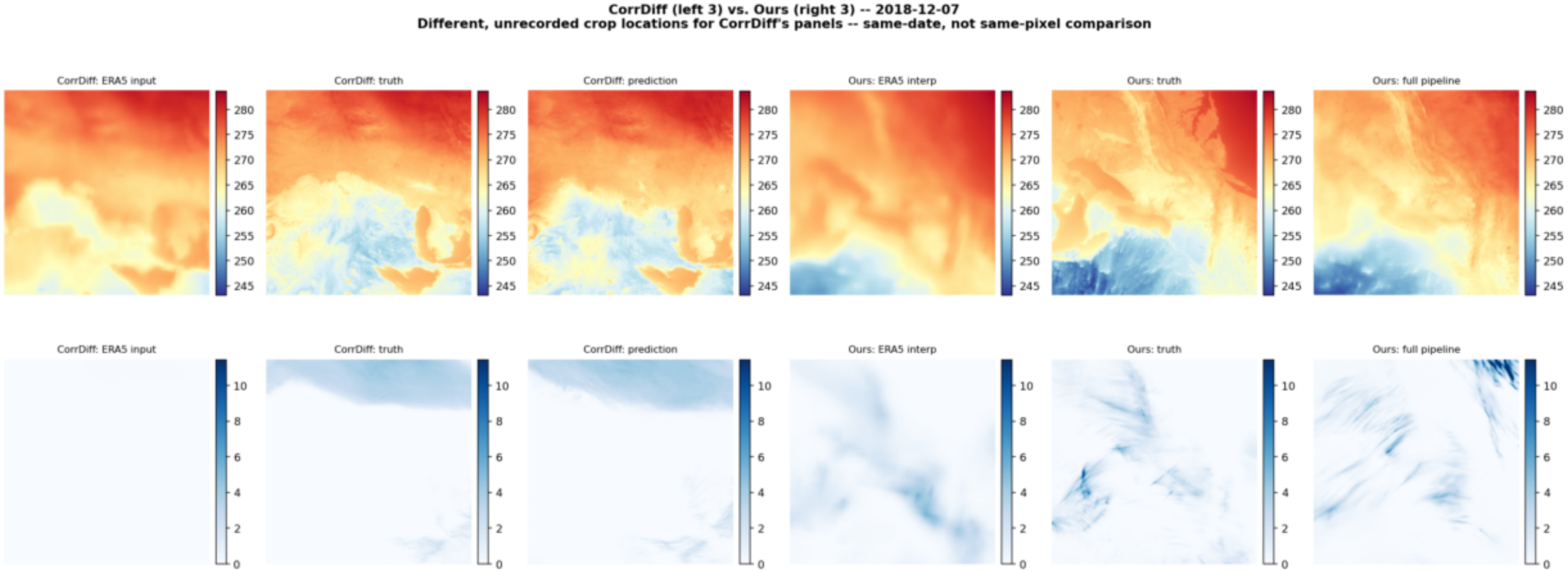

CorrDiff (official physicsnemo code), training time, one clean pass
GPU-hours
0
100
200
300
400
318.6
476.1
Regression
(5M steps)
Diffusion
(5M steps)


Our latent model (DRN + VAE + Diffusion), training time, clean path only
GPU-hours
600
500
400
300
200
100
0
326
149
582
DRN
VAE
Diffusion
(epoch 106/300)

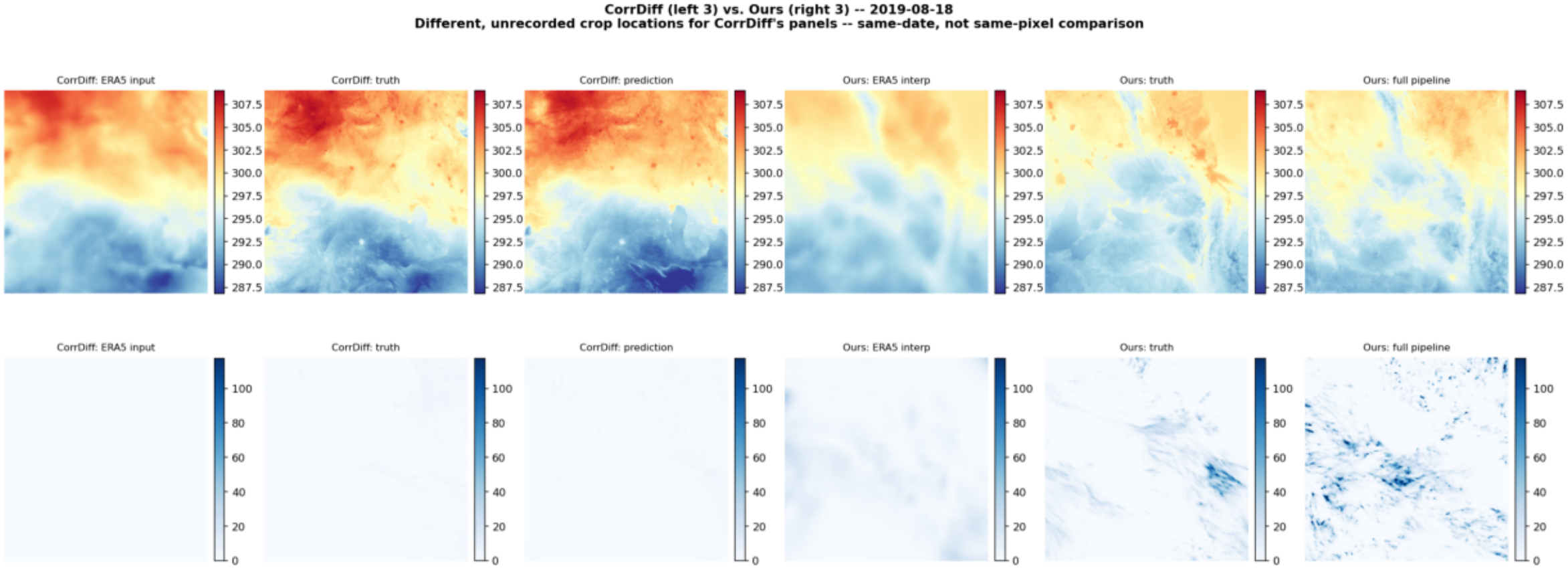
corrdiff vs ours, 2018-08-09, same date, different (unrecorded) crops



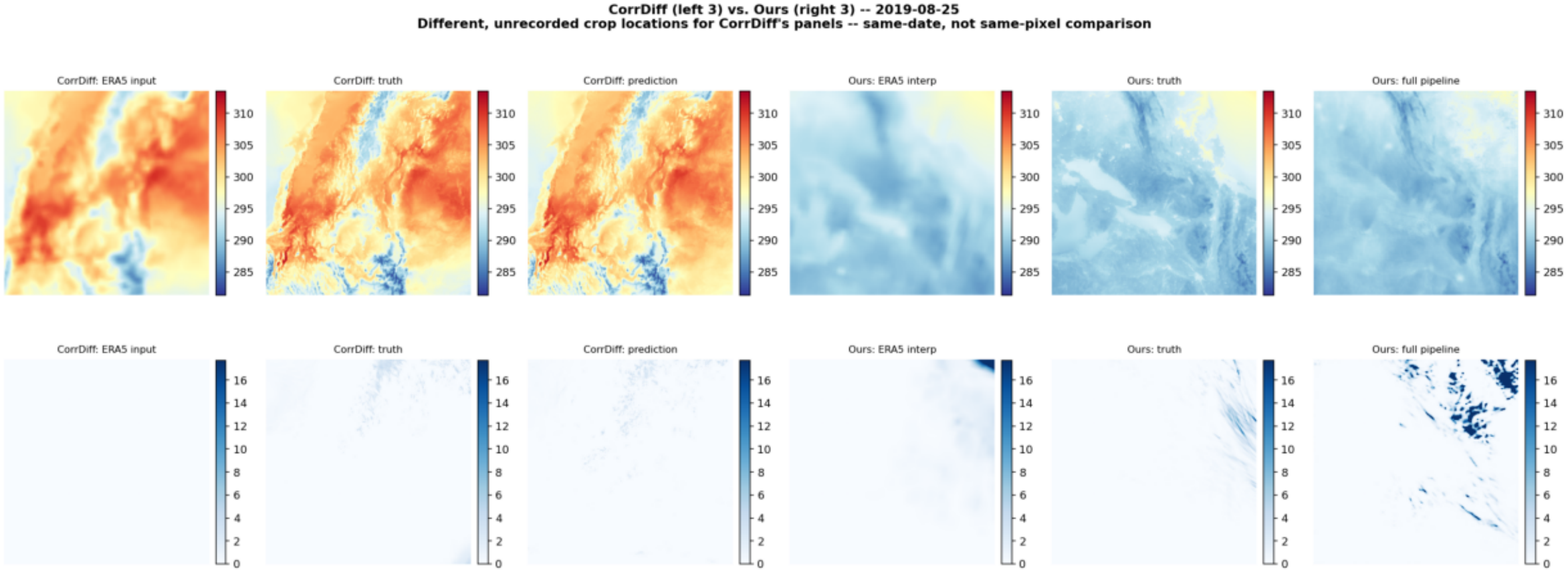
corrdiff vs ours, 2018-12-07, same date, different (unrecorded) crops



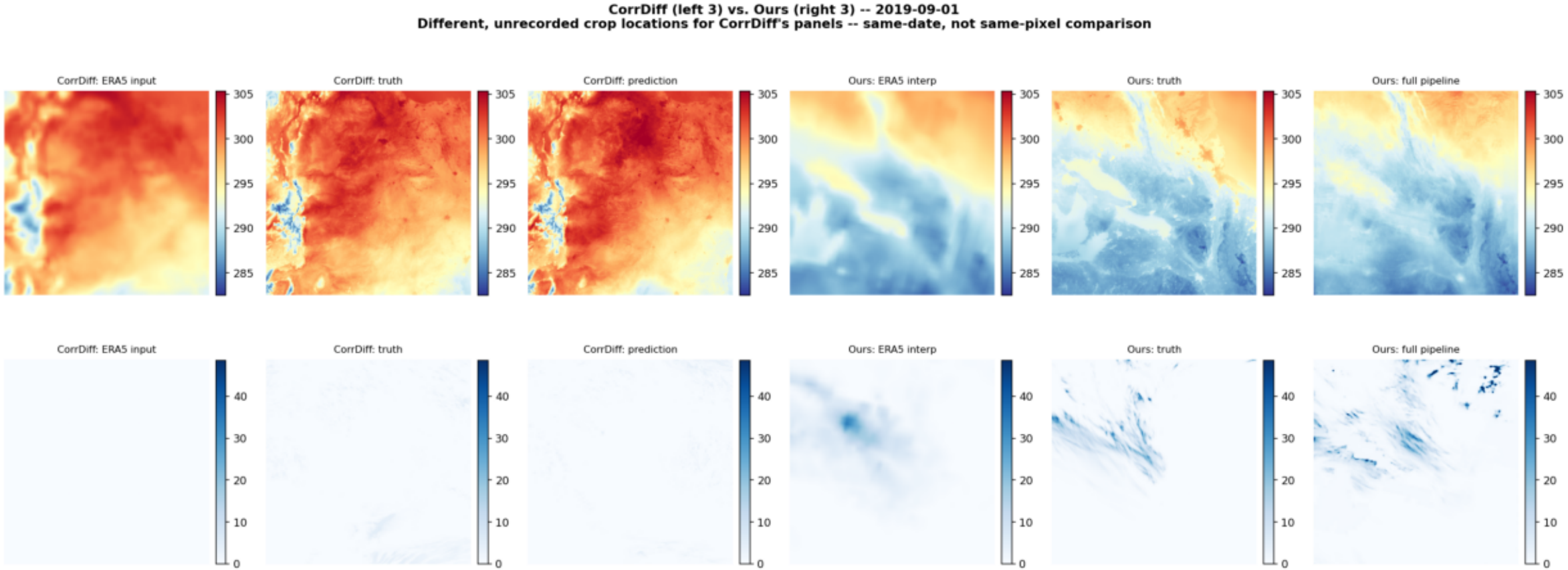
corrdiff vs ours, 2019-08-18, same date, different (unrecorded) crops



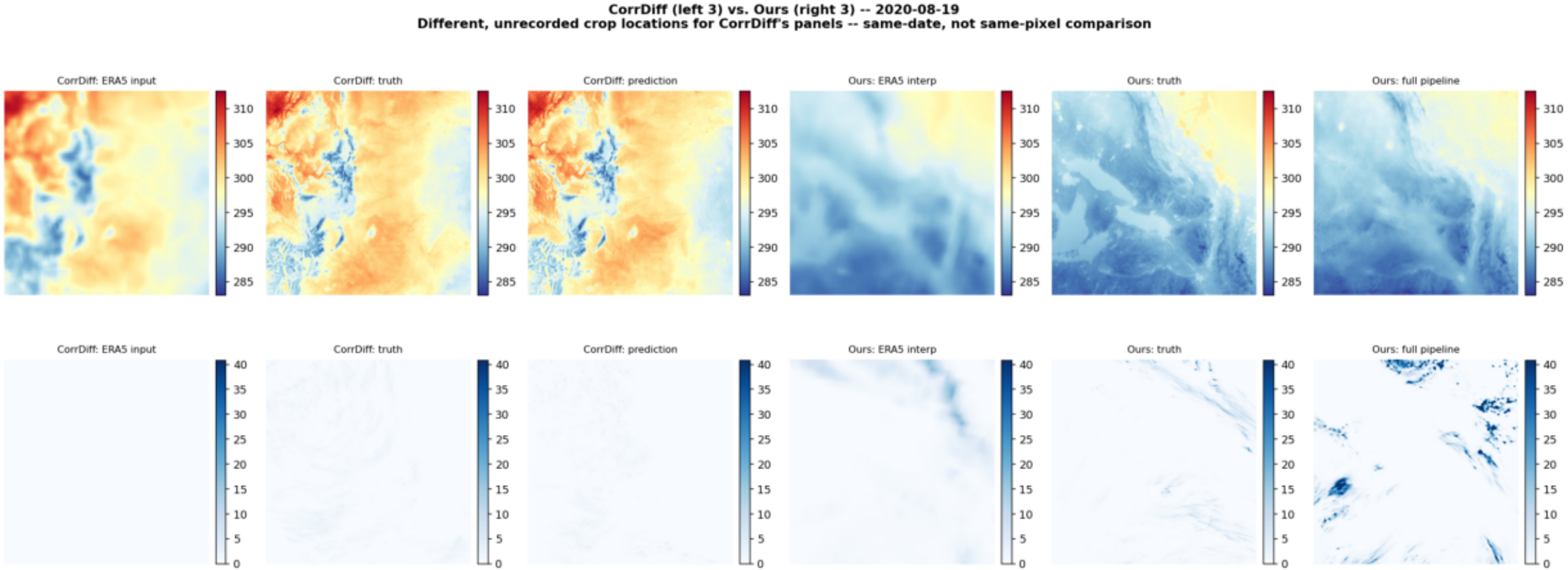
corrdiff vs ours, 2019-08-25, same date, different (unrecorded) crops



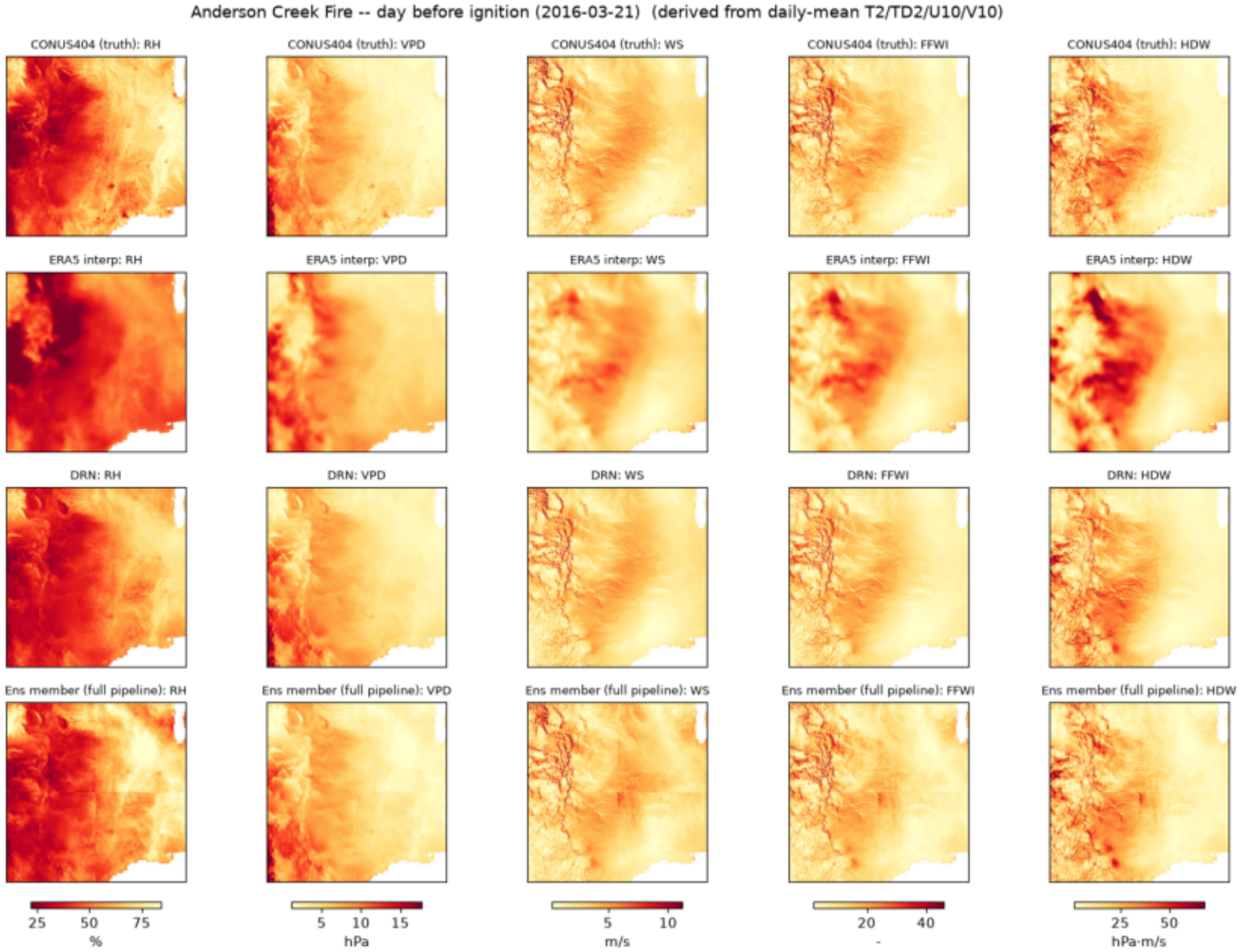
corrdiff vs ours, 2019-09-01, same date, different (unrecorded) crops



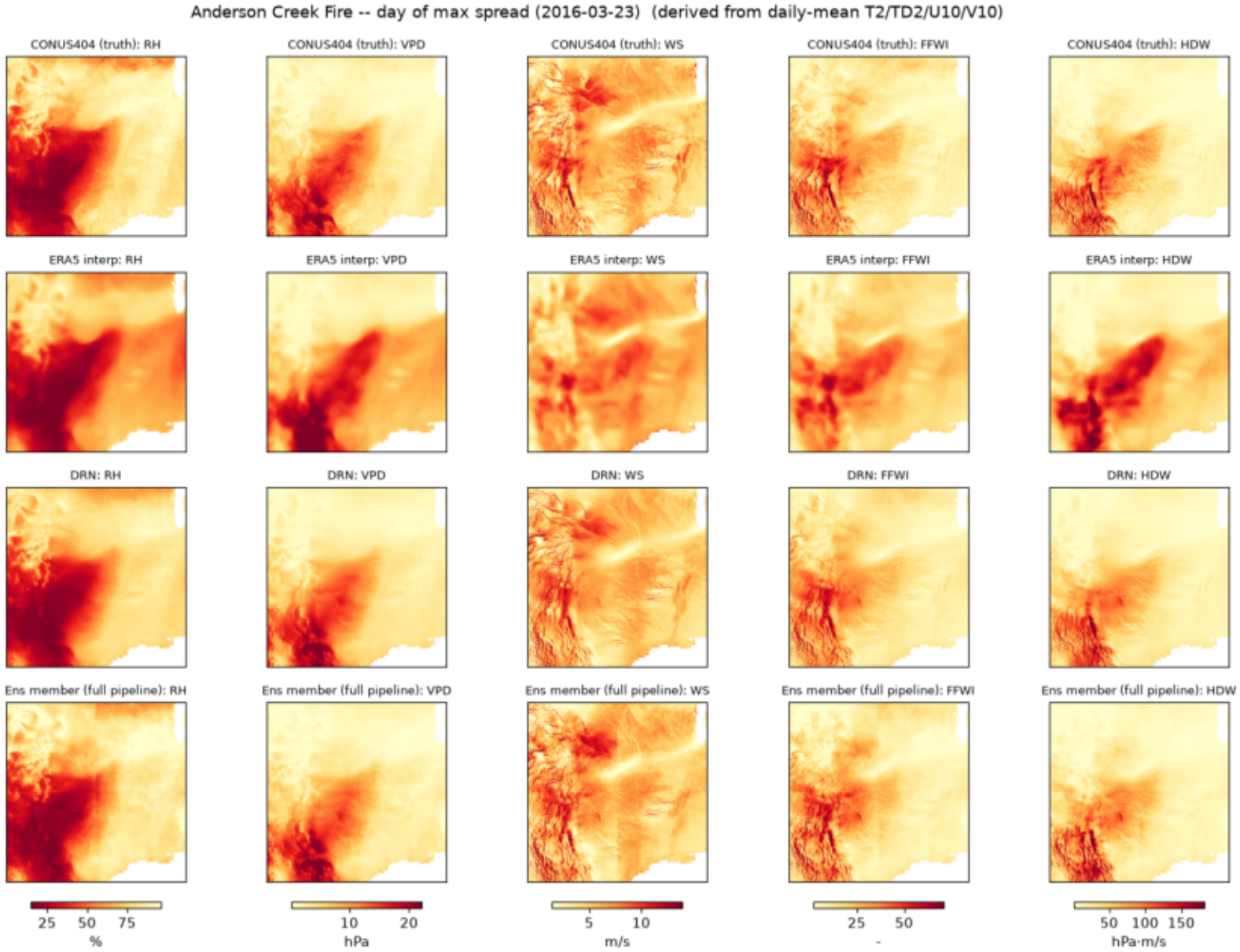
corrdiff vs ours, 2020-08-19, same date, different (unrecorded) crops



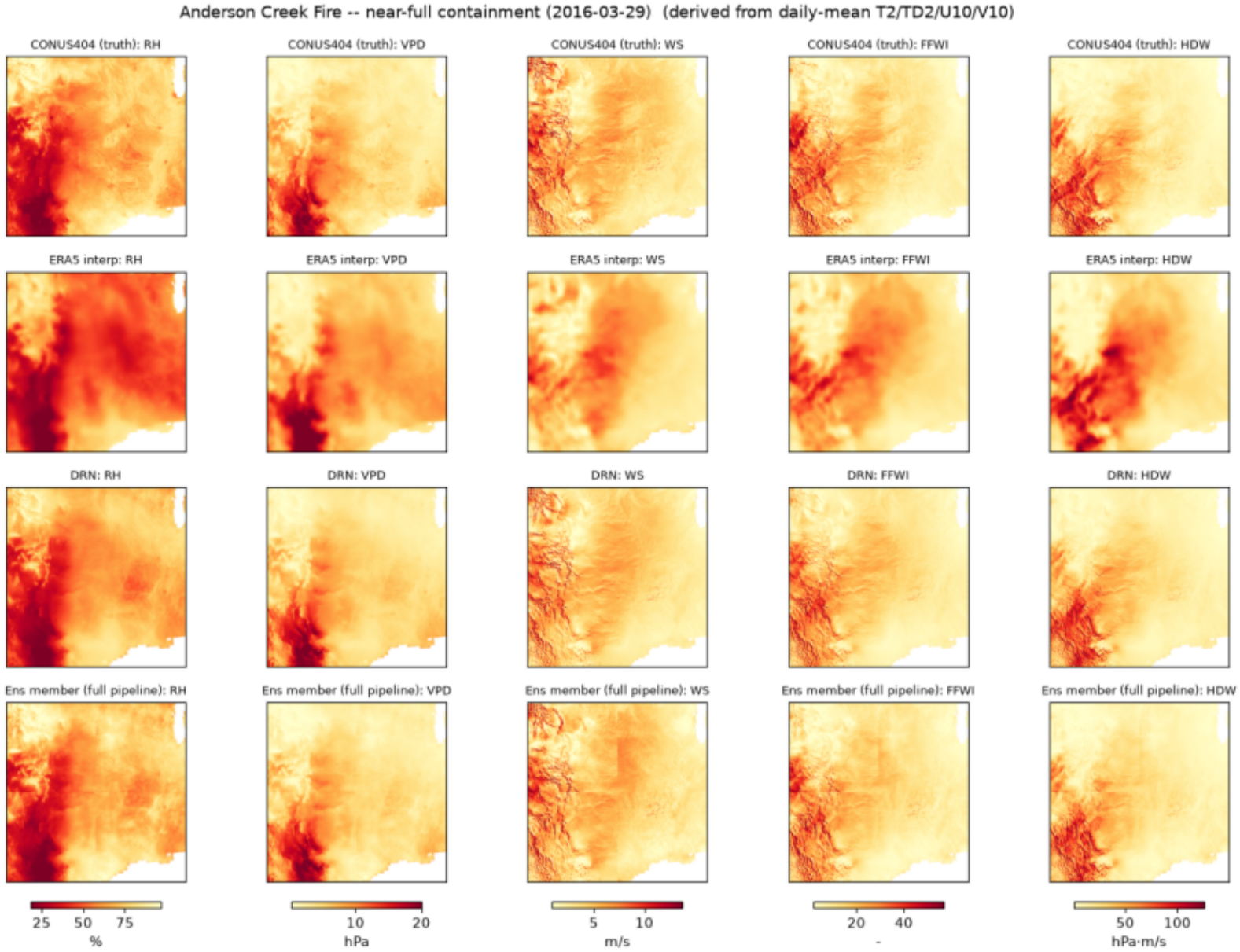
Anderson Creek Fire, day before ignition (2016-03-21)



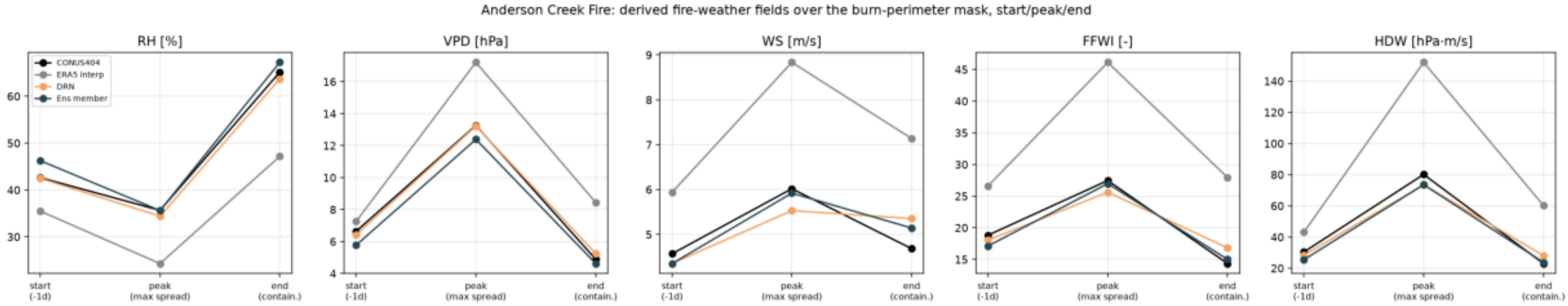
Anderson Creek Fire, day of maximum spread (2016-03-23)



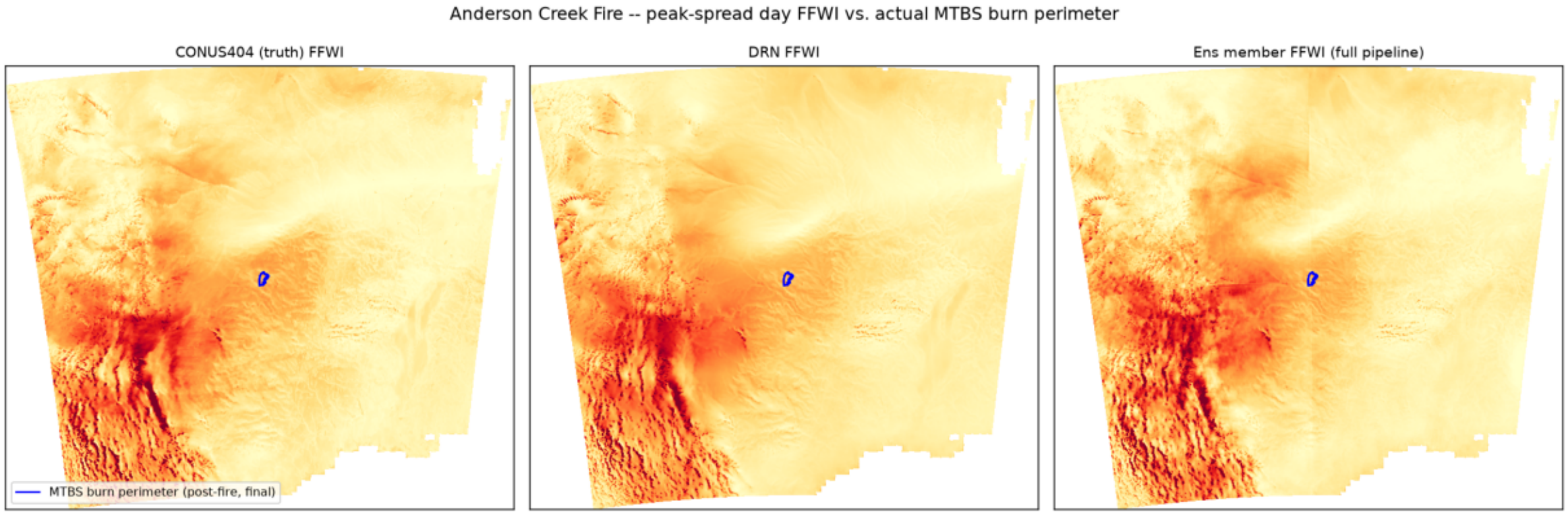
Anderson Creek Fire, near-full containment (2016-03-29)



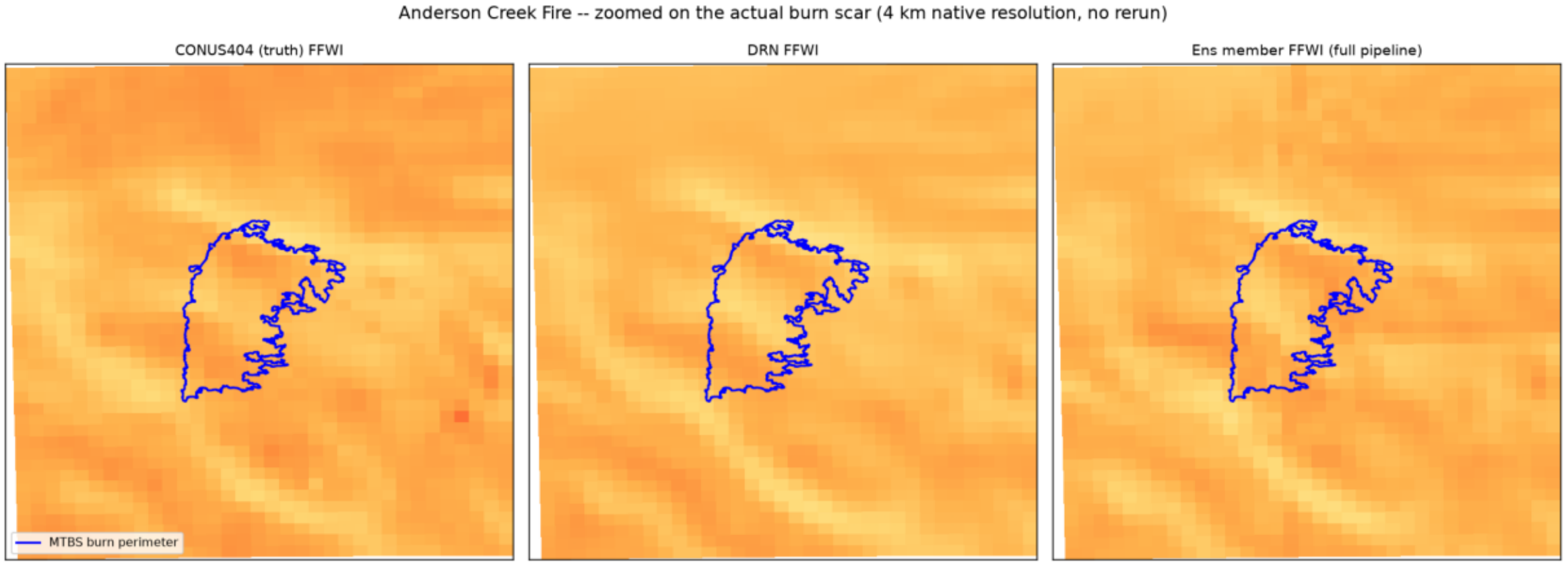
Fire-weather timeline: start → peak → containment



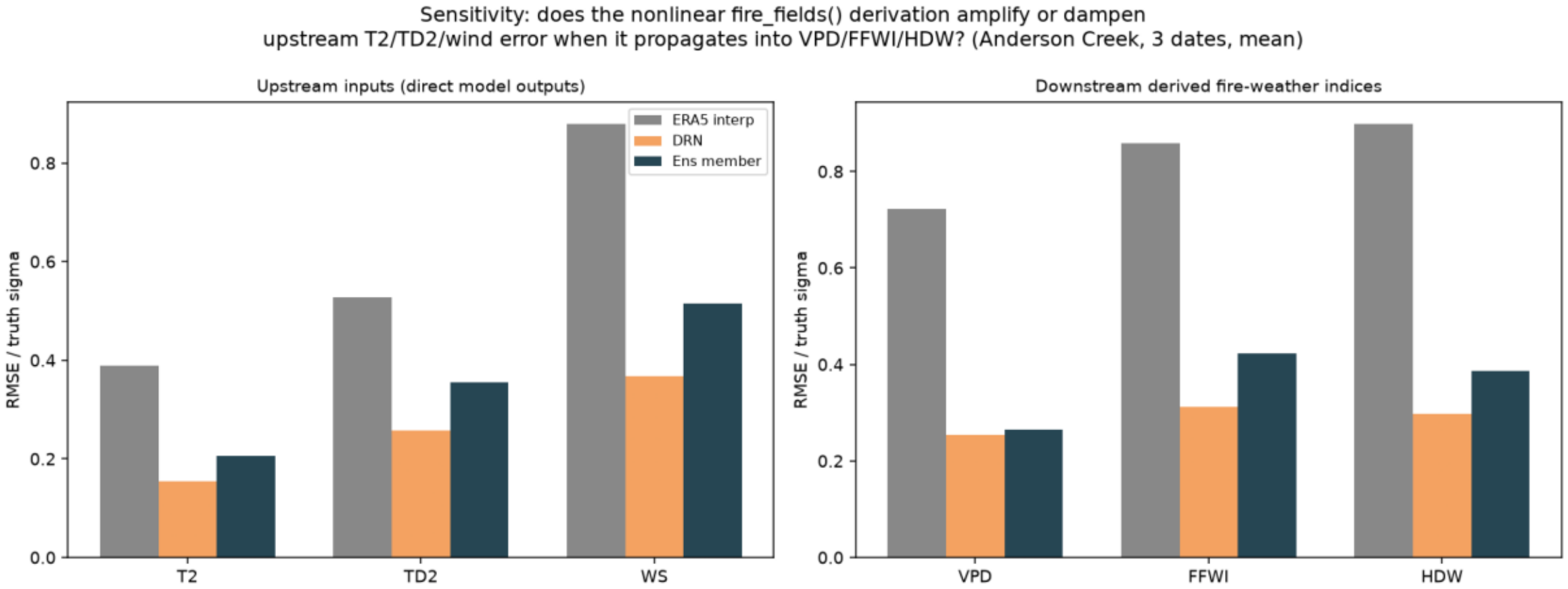
Peak-spread FFWI vs. the actual MTBS burn perimeter



Zoomed on the actual burn scar (4 km native resolution)



Sensitivity: does the fire-weather formula amplify upstream error?



Independent station check (AVK ASOS, ~46 km from the fire) vs. model

